

Machine Learning Algorithms

A visual framework — 10 algorithms, 2 paradigms, 4 domains

TYPE	● Supervised	● Unsupervised	DOMAINS	Tabular Data	Computer Vision	NLP	Generative AI
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SUPERVISED LEARNING — requires labeled training data

01 Supervised Linear Regression HOW IT WORKS Fits a straight line through labeled data points to predict a continuous numeric output. Minimizes total prediction error using least squares optimization. DOMAINS Tabular Data REAL-WORLD USE CASES Predicting house prices · Forecasting monthly sales · Estimating fuel efficiency	02 Supervised Logistic Regression HOW IT WORKS Despite the name, this is a classification algorithm. Uses an S-shaped sigmoid curve to estimate the probability an input belongs to a category, then assigns a label. DOMAINS Tabular Data · NLP REAL-WORLD USE CASES Email spam detection · Predicting loan defaults · Binary sentiment classification	03 Supervised Decision Tree HOW IT WORKS Splits labeled data into branches using yes/no questions about feature values. Each split creates the purest possible groups. Predictions are made at leaf nodes. DOMAINS Tabular Data REAL-WORLD USE CASES Medical diagnosis support · Customer churn prediction · Product routing logic	04 Supervised Random Forest HOW IT WORKS Builds many Decision Trees, each on a random data subset. Aggregates votes across all trees. Reduces the tendency of any single tree to overfit training data. DOMAINS Tabular Data · Computer Vision REAL-WORLD USE CASES Fraud detection in banking · Patient readmission risk · Credit scoring
05 Supervised Support Vector Machine HOW IT WORKS Finds the widest possible margin between classes. Handles nonlinear problems by projecting data into higher dimensions using kernel functions. DOMAINS Tabular Data · Computer Vision · NLP REAL-WORLD USE CASES Handwritten digit recognition · Text classification · Gene expression analysis	06 Supervised K-Nearest Neighbors HOW IT WORKS Classifies a new point by finding its K most similar labeled training examples and taking a majority vote. Similarity is measured by distance in feature space. DOMAINS Tabular Data · Computer Vision REAL-WORLD USE CASES Product recommendations · Anomaly detection in sensors · Image classification	07 Supervised Neural Network HOW IT WORKS Layers of connected nodes pass information forward with learnable weights. Adjusts all weights backward through backpropagation to reduce error across labeled examples. DOMAINS Tabular Data · Computer Vision · NLP · Generative AI REAL-WORLD USE CASES Autonomous vehicle vision · Voice assistants · Real-time language translation	

UNSUPERVISED LEARNING — finds structure without labels

08 Unsupervised K-Means Clustering HOW IT WORKS Places K centroids randomly, assigns each unlabeled point to its nearest centroid, then moves centroids to the average of their group. Repeats until no points change — convergence. DOMAINS Tabular Data · Computer Vision REAL-WORLD USE CASES Customer segmentation · Grouping satellite images by land type · Document topic discovery	09 Unsupervised Principal Component Analysis HOW IT WORKS Reduces the number of features by finding new directions — principal components — that capture the most variation. Compresses information while retaining what matters most. DOMAINS Tabular Data · Computer Vision REAL-WORLD USE CASES Compressing facial recognition features · Visualizing genomic data · Sensor noise reduction	10 Unsupervised Generative Adversarial Network HOW IT WORKS Two networks compete: a generator creates fake samples, a discriminator tries to detect them. Each improves in response to the other until outputs consistently fool the discriminator. DOMAINS Computer Vision · Generative AI REAL-WORLD USE CASES Generating synthetic faces · Creating rare medical training images · Artistic style transfer
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SUPERVISED — how these algorithms relate
→ All 7 require labeled training data — a known correct answer for every example
→ Linear and Logistic Regression are the simplest baselines for numeric and categorical outputs
→ Random Forest is built from many Decision Trees, correcting overfitting by aggregating votes
→ SVM and KNN both rely on distance geometry but scale very differently as dataset size grows
→ Neural Networks can approximate the behavior of most other supervised methods given enough data

UNSUPERVISED — how these algorithms relate
→ None require labels — structure is discovered from the data itself
→ PCA is commonly run before K-Means to reduce noise and improve cluster separation
→ K-Means finds groups; PCA finds dimensions — adjacent but distinct problems
→ GANs use an internal supervised signal (real vs. fake) to learn without external labels
→ All three can serve as preprocessing or augmentation tools for downstream supervised models

Dimension	Supervised	Unsupervised
Training data	Requires labeled examples with known correct answers	Works on raw unlabeled data — no correct answers needed
Goal	Predict a class label or numeric value	Discover hidden structure, patterns, or groupings
Evaluation	Accuracy, precision, recall against known labels	Cluster coherence or quality of generated output
Typical output	A prediction or classification for each new input	Cluster assignments, compressed features, or generated samples
Best used when	You have labeled data and a clear prediction target	You want to explore structure or lack labeled data

Domain coverage at a glance				
Algorithm	Tabular	Vision	NLP	GenAI
Linear Regression	✓			
Logistic Regression	✓		✓	
Decision Tree	✓			
Random Forest	✓	✓		
Support Vector Machine	✓	✓	✓	
K-Nearest Neighbors	✓	✓		
Neural Network	✓	✓	✓	✓
K-Means Clustering	✓	✓		
PCA	✓	✓		
GAN		✓		✓